The Timing of Early-Life Shocks and Adolescent Mental Health

October 2026

Do the effects of an earthquake experienced at ages 0–4 persist?



Shock 27 February 2010 (27-F), magnitude 8.8

Exposed Children aged 0–4

Effects of early-childhood shocks

Birth weight (Torche, 2011)

Cognitive and non-cognitive development (Gillmore, 2026)

Height and learning (Andrabi, Daniels, and Das, 2023)

Adult mental health (Persson and Rossin-Slater, 2018)

This paper

Mental health in adolescence, ages 14–18

Does the age at which the shock is experienced matter?

> "The effects of stress on the brain depend critically on the timing (age of onset and duration). When stress occurs early in life it can have profound and lasting effects on brain organization and function." (Gee and Casey, 2015, p. 184)

Potential mechanisms

	Channels
Socioeconomic and family environment	The earthquake reduces income, damages housing, and increases conflict at home (Conger and Donnellan, 2007; Almond, Currie, and Duque, 2018)
Transmission of maternal stress	Infants regulate their stress through the mother. If she is stressed, she passes it on through daily contact (Persson and Rossin-Slater, 2018)

Data



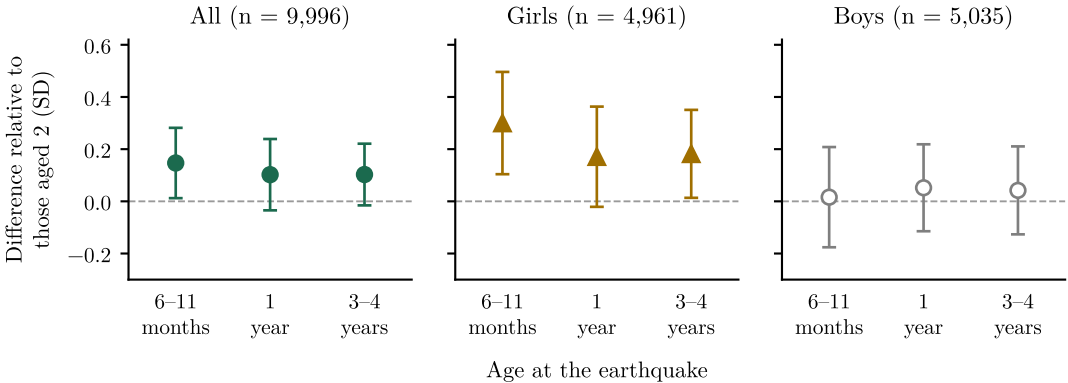
Survey	ELPI, Early Childhood Longitudinal Survey, Chile
Sample	9,996 adolescents born in 2006–2009; 4,961 girls
Respondents	The caregiver in every round; the child in 2017 and 2024

Econometric specification

$$Y_{imc} = \sum_{a \neq 2} \beta_a \left(\text{Age}_{ai} \times \text{Affected}_m \right) + \sum_{a \neq 2} \gamma_a \, \text{Age}_{ai} + \psi_m + \theta_c + X_i' \delta + \varepsilon_{imc}$$

Y_{imc}	Mental health at ages 14–18, PHQ-4
Age_{ai}	Age at the earthquake: 6–11 months, 1 year, or 3–4 years; reference, 2 years
Affected_m	Municipality in the zone of destructive damage
ψ_m, θ_c	Municipality and birth-cohort fixed effects
X_i	Child and household controls

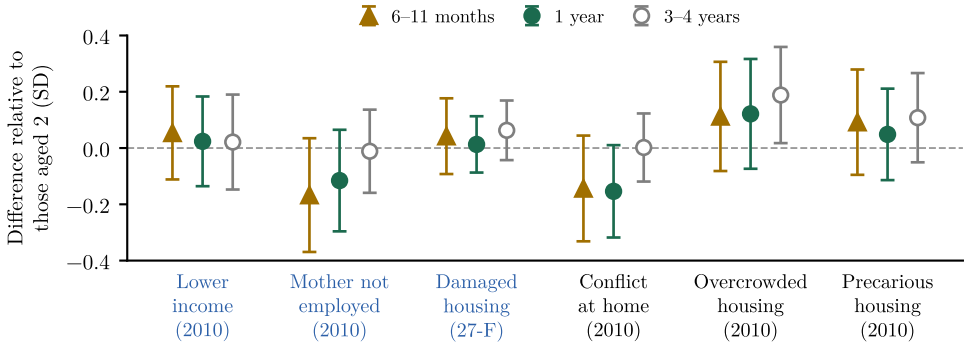
Effect on mental health



Mental health is measured with the PHQ-4 (Kroenke et al., 2009), four questions answered by the adolescent on symptoms of anxiety and depression (nervousness, worry, loss of interest, and feeling down). Each point is the difference between affected and unaffected municipalities, with 95% CI.

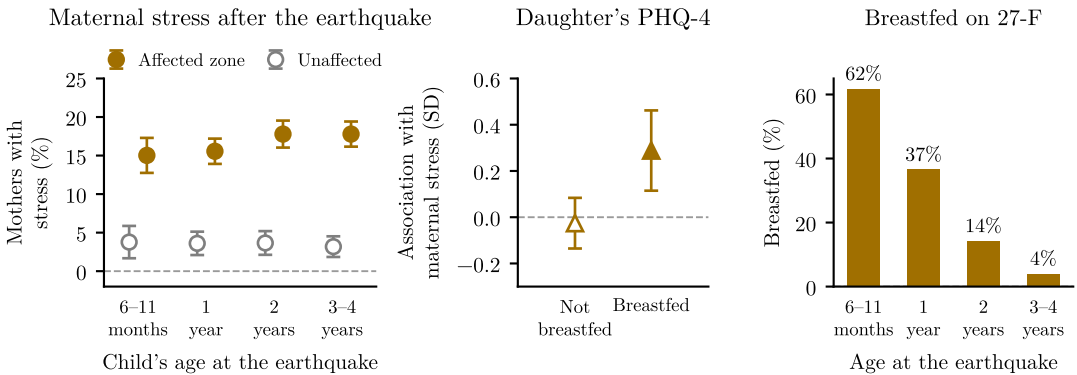
Socioeconomic and family environment does not explain the gap

The earthquake worsened the environment (Gillmore, 2026), but with no differences by age



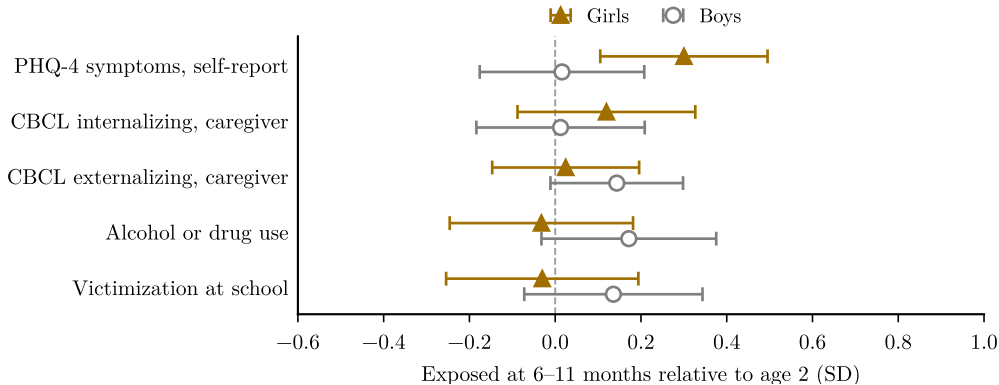
Measured in 2010, one to nine months after the earthquake. In blue, the mechanisms in Gillmore (2026). Lower income, per capita; damaged housing, destroyed or with major damage; conflict at home, the mother shouting, being hostile, or hitting during the visit; overcrowded housing, persons per bedroom; precarious housing, an emergency shack (*mediagua*) or a lent or shared dwelling. Girls, relative to those aged 2; 95% CI.

Maternal stress and daughters' mental health by breastfeeding



Active breastfeeding on 27-F serves as a proxy for regulatory dependence and biochemical and behavioral proximity to the mother. Girls, with fixed effects and controls, also for the mother's education and employment and for age; 95% CI.

Among boys, only weak signs in behavior and substance use



What the data suggest and what remains open

What the data suggest

- Girls exposed at 6–11 months have worse mental health than those exposed at age 2, by 0.30 SD on the PHQ-4.
- The socioeconomic and family environment changed similarly at all ages and does not explain this difference.
- Maternal stress is associated with worse mental health only among daughters who were breastfed, 62% of those exposed at 6–11 months.

What remains open

- What explains the effect at ages 3–4, if it operates neither through the socioeconomic and family environment nor through maternal stress?

  Hypothesis: at that age children can already remember consciously what they experience, and the memory of the earthquake could affect their mental health.